PROJECT CASE STUDY

HexaQuote

AI Voice-to-Quote React Native Platform

Category	Status	Year
Full-Stack AI	Full-stack prototype	2026

A tradesperson records a job description; Whisper transcribes it, Llama structures the costs, Supabase stores the quote, and FastAPI returns a professional PDF for sharing.

2	~0.7s	Offline
integrated services	documented transcription	queue + sync

Context and problem

Tradespeople often create estimates manually after a site visit, losing time converting spoken notes into structured, professional quotes.

Strategy and implementation

- Built an Expo/React Native frontend and FastAPI backend communicating over HTTP.
- Used Groq Whisper for transcription and Llama 3.1 for strict JSON extraction.
- Stored quotes in Supabase and generated PDFs with fpdf2.
- Added platform-aware uploads, recent-quote history, offline queues, and automatic synchronization.

My role

End-to-end product design, AI orchestration, backend APIs, mobile integration, database layer, PDF generation, and resilience behavior.

Technology stack

React Native | Expo | FastAPI | Uvicorn | Groq Whisper | Llama 3.1 | Supabase | fpdf2

Results and evidence

- Converted voice notes into validated quote objects and shareable PDFs.
- Separated frontend on port 8081 from backend on port 8000.
- Implemented offline recording queues and auto-sync when connectivity returns.

Key takeaways

- LLM output must be validated through a strict schema.
- Mobile networking and offline behavior are product requirements, not edge cases.

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